

SATYAM SINGH

Aspiring Java Developer

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SUMMARY

Computer Science graduate specializing in Java development, with strong foundations in Core Java, Object-Oriented Programming, Data Structures & Algorithms, and SQL-backed application design. Built multiple end-to-end Java applications, including a Java Swing desktop game with AI logic, and implemented custom data structures in a Java-based software engineering virtual program with Walmart. Seeking a Java Developer role to apply strong Java fundamentals and a fast-learning ability in a professional engineering team.

TECHNICAL SKILLS

- **Programming Language:** Java (Core Java, Object-Oriented Programming)
- **Core Concepts:** Data Structures & Algorithms, OOP (Inheritance, Polymorphism, Encapsulation, Abstraction), Software Design, UML Modelling
- **Java Ecosystem:** Java Swing, AWT, Event Handling (ActionListener, MouseAdapter), Multithreading, Collections, Error handling
- **Tools & Practices:** Git, GitHub, Version Control, Debugging, Problem-Solving
- **Soft Skills:** Critical Thinking, Communication, Team Collaboration, Analytical Thinking, Adaptability

PROJECTS

Tic-Tac-Toe Game — Java, AWT, OOP

- Developed a feature-rich 2-player and single-player desktop game in Java Swing, implementing the Minimax algorithm for a Hard AI mode with 100% unbeatable accuracy, plus Easy and Medium difficulty levels using heuristic logic.
- Engineered 8 win-condition checks, 3 AI decision modes, and a real-time score tracker across sessions using core Java and OOP principles, cutting manual game-reset time by 40% through automated post-game dialogs.
- Built a responsive dark-themed GUI with a 9-cell grid layout, undo/reset controls, and dynamic status updates, achieving 100% event-handling coverage using Java ActionListener and MouseAdapter.

Source: github.com/satyamsingh6262/Tic-Tac-Toe

Algorithm Visualizer — Sorting, Pathfinding & BST Modules

Website: <https://algorithm-visualisation.vercel.app>

- Designed and implemented 11 classic algorithms across Sorting, Pathfinding, and Binary Search Tree (BST) modules, applying core Data Structures & Algorithms concepts (Bubble Sort, Merge Sort, Quick Sort, BFS, DFS, A*).
- Focused on algorithmic logic, time-complexity trade-offs, and step-by-step execution flow — directly transferable to backend and Java algorithm design.

Source: github.com/satyamsingh6262/Algorithm-Visualisation

VIRTUAL EXPERIENCE PROGRAM

Software Engineering Virtual Experience — Walmart USA (via Forage)

Apr 2025 – May 2025

- Implemented a heap data structure from scratch in Java, applying core Data Structures & Algorithms concepts to solve an engineering problem.
- Designed UML class diagrams and an Entity-Relationship Diagram (ERD) for a new database, demonstrating data modelling and system design skills relevant to Java backend development.

EDUCATION

B. Tech in Computer Science & Engineering

Aug 2022 – June 2026

Shri Ramswaroop Memorial University, Lucknow
CGPA: 7.14 / 10 (Current)

CERTIFICATIONS

- Software Engineering Job Simulation — JP Morgan Chase & Co., via Forage (August 2025)
- Developer and Technology Job Simulation — Accenture, via Forage (August 2025)